



Colin Klein

WHAT THE BODY COMMANDS

The Imperative Theory of Pain

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For Paul R.

Physical pain is thus the psychical adjunct of an imperative protective reflex.
–Sherrington, The Integrative Action of the Nervous System.

In pain I hear the captain's command: "Take in the sails!"
–Nietzsche, The Gay Science §318

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1.1 Imperativism

I have a pain in my ankle—a dull ache, present intermittently for the past few weeks. Before that, there was a hot, throbbing pain for a few days. And before *that*, there was a brief twinge as my ankle rolled on the pitch.

Despite these variations in quality, intensity, and duration, each of these sensations was a pain. In virtue of being pains, they have some common feature. They share that common feature with sensations I have felt in a wide variety of other situations: when splattered with hot grease, cut shaving, eating spicy Sichuan food, bringing my hand toward a campfire, wrenching my wrist, and on and on. That common feature distinguishes pains from other sensations like the sight of a cricket ball, the smell of woodsmoke, or my hunger for bacon.

In what follows, I will give an account of that common feature. I will argue that pains are *imperatives*.¹ They are sensations with a content, and that content is a *command* to protect a part of your body. Typically, protection will involve limiting activity: the ache in my ankle tells me to avoid putting weight on it, the throb of a burn to protect my skin, and so on. All pains have imperative content, and that imperative content is what distinguishes them as pains.

Call this *imperativism* about pain.² Imperativism offers a compelling story about two otherwise puzzling features of pain. First, pains are strongly and directly *motivating*. The pain in my ankle motivates me to avoid walking if I can and to walk carefully if I do. Pains motivate *directly*: when I feel pain, I don't need to deliberate further about whether I am moved by it. To be in pain just *is* to be motivated. Pains' motivation is similarly *unmediated*: my pain motivates me regardless of my desires, values, or other motivational states. Indeed, pain is usually at odds with my desires (I'd really *like* to walk around normally, and but for the pain I would). Finally, pains give us *reasons* to act rather than just pushing us around. I might have other, better reasons to walk, reasons that override the reasons my pain gives me not to. Those reasons must be in addition to, and stronger than, the reasons provided by pain. Even if I were to become convinced that my ankle were fine, and that walking on it posed no threat, my pain would *still* give me a strong reason not to walk.

1 I have argued this in several places over the years. See especially Klein (2007, 2010, 2012).

2 I believe the name “imperativism” is originally due to David Bain (2011).

Motivation is so tightly connected to pain that it's hard to imagine it absent. As Hall (2008) notes, if someone claimed to feel pain but be completely unmotivated by it, we would ordinarily doubt that they were in pain at all. Philosophers may be familiar with well-known pathological cases where precisely this seems to happen, of course. I'll deal with these in chapter 11. For now, note that such cases are at least *strange* and demand explanation. They fascinate precisely because ordinary life offers nothing like them: usually there is no gap between pain and motivation.

Second, pains are remarkably *uninformative*. Although the pain in my ankle motivates me, it gives me few clues as to why it's actually there. I know, of course, that I sprained my ankle, and that the sprain causes my pain. That is not information carried *by* the pain, however: I only know my ankle was sprained because I turned it a bit, and the next day it looked like an angry grapefruit. That's good evidence that I sprained it. But I could be wrong. My physician took time to rule out alternative causes, and that was not an absurd thing to do. When I can't infer causes, I often have no idea why I'm feeling pain: a mysterious backache, say, gives no indication *whatsoever* about what is causing it. Pains are uninformative partly because their potential causes are so diverse. Some pains are caused by *actual* tissue damage: I feel a pain when, and because, the hot grease hits my arm. Other pains are caused by merely *potential* damage: the hand as it approaches the fire, the needle pressed just shy of puncture, the muscle stretched to its limit. Note that potential damage need not be *imminent* damage. Sitting in an odd posture causes pain, but I'd have to do that repeatedly and for a long time before I did any harm.

Pains of *exertion* have an even more tenuous relation to potential damage. Running causes my thighs to burn, and bench-presses make my pectoral muscles ache. But I probably can't run far enough to injure my legs, and bench-pressing hurts at weights well shy of my maximum ability.

Finally, there are pains of *recuperation*, which are associated with *past* damage. For all I know, the actual damage to my ankle has mostly healed. It remains inflamed and sensitive. That is probably a good thing. Although healed, the ankle is still weak. I am in danger of re-injury if I use it too vigorously. Pains of recuperation form the vast bulk of the pains we feel. In chapter 3, following P.D. Wall, I'll suggest that pains of recuperation are also the biologically essential ones. Imperativism is crafted in part to place pains of recuperation front and center.

These two features of pain—its strong motivating power, on the one hand, and its uninformative nature, on the other—distinguish it from sensations in

other modalities. The sound of a magpie doesn't motivate me on its own. I often hear a magpie and do nothing. When its cry does motivate, what it motivates depends in complex ways on my other mental states. Sometimes the sound of a magpie causes me to look up, other times to duck, and still others to reach for the tennis racket. What I do depends on what I want and what I believe. Ordinary sensations are also informative about the world. Under ordinary conditions, the sound of a magpie informs me that there's a magpie about. I might be wrong about that; the link is not infallible. But the whole *point* of auditory (or visual, tactile, etc.) sensations is to inform me about the world.

Hence the puzzle. Ordinary sensations inform but don't necessarily motivate. Pains motivate without informing. That is why pain is unusual. A good theory about pain's nature should remove some of that puzzlement. I offer imperativism as that story.

Parallels with ordinary language imperatives are useful, and I'll draw them frequently. Ordinary imperatives also motivate without informing. If I say to you, "Shut the door!", I do so to get you to act. If you accept me as a legitimate source of commands, then my imperative will motivate you and give you a reason to close the door. My imperative doesn't convey any information about why I want you to close the door, it doesn't tell you anything about what the world is like (except indirectly), and its function is not to inform you. That's in part because commands aren't truth-apt: my command may be many things, but it is neither true nor false. The same is true of pains. That means there is at least a *prima facie* parallel between pains and commands. The rest of this book will cash out that parallel and show the philosophical benefits of doing so.

What I have described is not the ordinary view of pain. The ordinary view, I take it, is that pains inform about some bad bodily condition. Pain *is* like vision or hearing, except that it tells you about bodily damage instead of properties in the world beyond. Because that sort of information doesn't seem like enough to get you moving, motivation is handled by an additional appended component—an affective state, a desire, an evaluation, or something along those lines. Theorists debate just what the two components are, what their ontological status is, and whether both are strictly necessary for something to be called a pain.

I reject the ordinary view. I think typical presentations confuse aspects of pain proper with states evoked by pain. I think imperativism can do with one simple component what the ordinary view does with two complex ones. But most of all, I think the ordinary view is wrong about what pains are *for*. The biological role of pain is a homeostatic one. Like hunger or thirst, pain is there

to get you to act in ways that bring your body back into balance. Returning to balance only requires taking the right sort of actions. Your body doesn't need to tell you *why*—that information would only get in the way.

I will defend my picture at length in the chapters to come. As a brief illustration, however, consider my ankle. It aches. That aching keeps me from walking on it. I'm not sure why it aches. Maybe something is still torn down there, maybe it's just weak and needs special care. I don't know. Nor does it matter—whatever the reason, I should not walk on my ankle. If I can manage that, whatever is wrong will sort itself out. If I do walk on my ankle, it could make things worse, maybe permanently. Rather than leave it up to me, my body just motivates me not to walk around if I don't have to. Then I will heal. That is the function of pain.

This is why I will focus on pains of recovery. Pains that accompany recovery are the most prevalent, and arguably the most important, of the pains we feel. Philosophers often focus on pinpricks and other brief, acute pains. This has always struck me as slightly odd. The pain of a single sprained ankle will last longer than the total duration of all the pinpricks I will ever feel. Ignoring a pinprick has few consequences, and the threat is often gone before I notice. The pains of recovery, in contrast, are clearly adaptive and shape our actions for an extended time. It is obvious why they're there and obvious what they do. Focus on those, and imperativism already becomes more attractive.

1.2 Reflections on Imperatives

Imperativism sets up a parallel between pains and imperatives in ordinary language. I mean this quite literally: pains have the same kind of content, and play the same kind of role, as imperatives do.

A potential misunderstanding is worth cutting off at the outset. When I express the content of pain imperatives in English, I assume that the imperatives are instantiated in some form of mentalese, not natural language. Differentiating such imperatives from (say) *thoughts* with the same content is an interesting and complex project. I note that it's also one that *any* intentionalist theory faces, and so it has been rather widely discussed.³ I will say more about this in chapter 3; for now, as long as you think that sensory content can be

3 For a good review of possible responses, see Egan (2006). I'm most fond of solutions on which the imperative content of pain is sufficiently complex to distinguish it from other sorts of mental content. That's not implausible: the imperatives of pain proscribe against a wide range of complex

suitably distinguished from other sorts of mental content, imperativism raises no new or distinct problems.

Philosophers do not work with imperatives as much as they used to. That's partly due to changes in the profession. Thinking about imperatives has a rich philosophical history, but the rise of deontic logics alongside the troubles with developing imperative logics have made imperatives less attractive as a technical device.⁴

Yet imperatives play an important role in many areas of life. Parents regularly command children. Parenting techniques have moved away from an emphasis on obedience, but sometimes there is no substitute for a well-timed "Stop!" Formalized structures of commands play an important role in the military, of course, and similar hierarchical institutions. Informal commands can also motivate us, as when the trainer exhorts us to do one more lift. Commands don't require hierarchy, however. Two people can both be willing to accept commands from the other, and that willingness is crucial for regulating various kinds of joint projects. If you and I are moving a couch in a cramped house, then our exchange might consist entirely of commands to one another: "Higher!" "Drop your end!" "Now turn!" Similarly so for giving directions, or baking together, or all manner of mutually enjoyable activities.

Start with moving a couch: this strips away many of the complexities that arise when there are hierarchical relationships. Why are commands so natural? Conversely, why is it frustrating to move a couch with someone who chatters away about how they would like to move a couch, or what one should do in this situation, or what they'd prefer as far as couch-moving is concerned?

The answer, I suggest, is that imperatives are directly and nondeliberatively action-guiding. When I say, "Lift your end!" and all goes well, you just lift it. You don't need to deliberate about what would be best or right or good or what I might want. If you're committed to accepting my command, my command is enough to get you to act. We save time, and we act smoothly. Coordinated action in cases where there's good mutual information works best when there's as little deliberation as possible. Indeed, Lewis suggests that the distinction

ways of bodily action and so differ from thought imperatives just in virtue of their additional content. For a solution crafted especially for imperativism, see Martínez (2010).

4 Although not entirely. See, for example, Vranas (2008, 2010) and Parsons (2012a) for recent work in philosophy and Portner (2007) in linguistics. Hamblin (1987) is an excellent (although opinionated) introduction to the history of philosophical treatments of imperatives.

between indicatives and imperatives is that the former require deliberation on the part of the receiver whereas the latter don't.⁵

It is precisely this direct link between imperatives and actions that makes imperatives so efficient and useful. This is why they are common in hierarchical situations, where the deliberation of the agent is presumed to be precluded by that of his superiors. The extreme case of this, suggests Raz (1986), are the commands issued by societies in the form of laws. Laws are commands meant to encapsulate the deliberative wisdom of the state—and thereby preclude further deliberation on the part of citizens.

Of course, commands don't always have their desired effect. You might reject my authority. Commands can be as confusing and ambiguous as any other form of communication. You might treat my command as merely an expression of my preference, a threat, or a warning. From there, you might deliberate about what you should do.⁶

I take it, however, that these are all *failures* of command: if I balk, if I stop to deliberate, I have ceased to treat the command *as a command*. Contrast this with cases where an imperative fails because I can't carry out what you've ordered, circumstances are unforgiving, or I'm in the grip of some other, more pressing command. There the connection between command and action is preserved but blocked. There you *have* successfully commanded, and, had other things been easier, I would have done as you said. In such cases, I am motivated, but that motivation is overridden.

In ordinary cases, then, the function of imperatives is to provoke action rather than deliberation. That is why commands are not truth-apt: what is commanded is to be *made* true, and the interesting division is between commands that have been satisfied and those that haven't. When I flesh out the semantics of commands in chapter 5, I'll thus focus on *satisfaction conditions* rather than truth conditions.

1.3 Two Background Commitments

I am a pure imperativist, and I am an intentionalist about phenomenal character. Both commitments make imperativism harder to defend and so constrain and shape my account.

5 See Lewis (1969, 42ff) and discussion in Huttegger (2007).

6 On general considerations against the *reduction* of commands to threats or other legitimacy-carrying representations, see Parsons (2012a).

First, I defend *pure* imperativism. The pure imperativist claims that pains *only* command. They have no (psychologically relevant) content beyond their imperative content. Contrast this with *hybrid* imperativism, which claims that pains have additional content over and above what they command. Hybrid imperativism could take pains to be simple conjunctions of commands and representations, or they could appeal to more complex contents that both motivate and inform.⁷ As I write, imperativists are rare, and pure imperativists rarer still. Hall (2008) and Martínez (2010) both defend imperativism. Hall is explicitly a hybrid imperativist of the first sort about pain, whereas Martínez suggests the second route.

Second, I will assume that some form of *intentionalism* about phenomenal character is true. Intentionalism is, minimally, the claim that the phenomenal character of an experience supervenes on its intentional content (Byrne, 2001). This means that there can be no change in phenomenal character without a change in intentional content.

Intentionalism is often run together with *representationalism*, the claim that phenomenal content supervenes on how a state represents or describes the world. I accept that representationalism is true for some kinds of mental states. Sensations in visual modalities, say, have *indicative* or *representational* content. These modalities depict the world as being a certain way, and sensations in these modalities are veridical or nonveridical just in case the world is that way.

Representing the world is only one way of having content. Just as sentences can be indicative, imperative, or subjunctive, so mental states might come bearing a variety of different types of content. Imperativism simply claims that the subvening content in the case of pain is a *command* rather than a proposition. Here the contrast with vision becomes obvious. While vision says that the world *is* a certain way, pain tells you to *make* it a certain way. Although both have content, they have a different direction of fit. Vision has the sort of content that can be true or false. Pains have the sort of content that you can satisfy or fail to satisfy. Hence, the difference.

7 For example, along the lines of Millikan's pushmi-pullyu representations or Clark's action-oriented representations. Once upon a time I was more keen on these complex contents and defended them in early drafts of my (2007). Tim Bayne (2010a), although not explicitly concerned with pain, suggests something like a complex hybrid account in passing. Manolo Martínez has also indicated to me in conversation that he's attracted by this line, given the utility of Millikan's account in other areas.

Intentionalism is controversial. I am not interested in defending intentionalism (except perhaps indirectly by neutralizing a problem case). I think intentionalism is the best theory going for how to naturalize phenomenal character—I don't see how *else* you'd manage. As a naturalist, I suppose I'll stick with the most promising option. In any case, my adoption of intentionalism should not be objectionable given my aims. For one, phenomenal character is mysterious. Rather than remain mute and passive in the face of that mystery, we might as well soldier on and describe what we can describe in the best way we can. Content is a good place to start, and we can say a lot of interesting things about the content of pain without solving that deeper mystery.

What's more, the combination of intentionalism and pure imperativism makes my job much harder than it would be otherwise. The two together restrict my explanatory resources down to a single source: the imperatives that constitute pain. I will have to use that one resource to do a lot of things.

I have mentioned, for example, that pains are distinguished from ordinary sensations by their imperative content. But this is not the only distinction that needs to be drawn. Pains are not the only imperative sensations. I think sensations like hunger, thirst, and itches also admit of an imperativist treatment. Each are sensations that motivate rather than inform. They are sensations that play a crucial role in keeping us alive and intact, and they do so precisely by commanding us to act in appropriate ways. Although I will have less to say about these imperative sensations, I must still tell a story about what distinguishes them from pains.

Pains also differ from one another. So while every pain has some character in virtue of which it counts as a pain, different pains are distinguished by their more specific features. A good theory should account for that variation as well. There are three important phenomenological axes along which pains vary. First, *location*. The pain in my ankle is distinguished from the pain in my wrist by its felt location. Location has a clear connection to motivation: the pain in my ankle motivates me to protect my ankle, not anything else. Protecting my ankle might have implications for what I do with other parts of my body (to hobble, I also have to move many muscles in unusual ways). These other actions are secondary to, and in service of, protecting my ankle. Second, *intensity*. The pain in my ankle is strong, whereas the pain in my wrist is mild. The intensity of pains has a clear connection to motivation as well: strong pains motivate me to take action more vigorously than weak ones do. Third, *quality*. Pains can be sharp, burning, wrenching, or aching. The McGill pain questionnaire, a standard instrument for rating pain, contains seventy-seven

different descriptors of this sort across a variety of categories (Melzack, 1983). I will argue that pain quality depends on *how* you should protect something: what differentiates an ache from a stab is the fine-grained details of the activity that's commanded. As a pure imperativist, I must argue that each of these differences between pains can ultimately be located in differences in the content of the command expressed by pains.

The above is just a sketch; later chapters will flesh out the details. For now, note that many of these problems are much *easier* if you are willing to abandon either intentionalism or pure imperativism. The pure imperativist has to grapple with problems that additional representational content solves trivially. For example, felt location takes a bit of work for a pure imperativist because commands aren't located. Adding representational content would make things easy—you can just locate the pain where whatever's represented happens to be. Similarly, accounting for pain's motivational power is easier if you relax the intentionalism and let yourself appeal to broader functional roles.⁸

Nevertheless, I think the combination of intentionalism and pure imperativism is worth defending. For one, I think it can be made to work, and that's interesting in its own right. Even if you end up thinking that pure imperativism fails, hopefully I can convince you that it has fewer shortcomings than you might have thought. Any additional content would thus play a smaller role. By trying to make pure imperativism work, and doing so in a way that is consistent with intentionalism, I intend to show how philosophically powerful an appeal to imperative content can be. That's why restricted accounts can be of philosophical interest.

1.4 The Plan of the Book

The goal of this book will be to defend pure imperativism. Because it is a relatively new account, I take my primary charge simply to be to show that imperativism is a *coherent* picture. That goal will take a fair bit of work: various objections have already been raised to it, and I've come to think that imperativism works best when presented as a whole package of views rather than piecemeal. Presenting that package, however, also requires motivating imperativism and giving some arguments in its favor.

8 Perhaps by adopting some form of quasi-intentionalism. See Ganson and Bronner (2013) for that sort of argument.

Most work on pain assumes that pain serves as a *symptom* of damage. I know that my ankle is sprained because of the pain, and that pain (somehow) gets me to act. In that sense, I am on an epistemological par with my doctor: she also knows that my ankle is sprained because I am in pain. My access to this symptom is a bit more direct than hers: unlike with the swelling, she can't just look and see. As far as what my pain signifies, however, and what should be done about it, the doctor and I are on a par.

I think that's absurd. If pain worked like that, then we wouldn't last long. The doctor can ignore my pains, can encourage me to walk despite the pain, and so on. The link between symptom and proposed action could be tenuous and arbitrary. My pain stands in a more intimate relationship to me and my ankle. It doesn't simply provide one piece of a riddle that I then solve on my own. Rather, my pain motivates me to do something about the problem. That's the biological advantage of feeling pain (and imperative sensations more generally): pain forces us to do the right thing. So here's the story in a phrase: in ordinary cases, pain is not a *symptom* but *part of the cure*.

To motivate pure imperativism, then, I'll start by thinking about the biological role of pain. On this model, pains belong to a class I will call the *homeostatic sensations*: sensations that motivate action to preserve some bodily parameters within acceptable limits. Others of this class include thirst, hunger, itch, and dyspnea (the felt need to breathe). Each of these sensations motivates rather than informs. I will present a model of imperative content which shows that such intrinsically motivating sensations are best understood as imperatives (chapter 2). Then I will argue that pain belongs to this class (chapter 3).

Doing so requires dealing with a fundamental ambiguity in the term "pain" as it is used in ordinary English. R.M. Hare (1964) suggests that there are three distinct uses of the word "pain."⁹ First, "pain" is used (as I will use it) to refer to the bare sensation that is common to pinpricks, postural adjustments, sprained ankles, and the like. This is the sense that I care about and the sense in which pain is a homeostatic sensation. This first sense carries no implication that the pain actually *hurts*, however. Indeed, I'll argue that there are common, nonpathological cases of pains that don't hurt or feel bad (just as there are hungers that don't hurt or feel bad).

Second, "pain" is used to refer to that sensation *plus* the accompanying felt badness. Third, some words pick out sensations that *hurt* but don't involve any pain in the first sense: this is what we mean when we talk about the pain of

9 See pages 94–5. I depart slightly from Hare's specific formulations.

heartbreak, the anguish of grief, and so on. As a paradigmatic case of the third use, consider Keegan's (2011) description of the English troops at Agincourt:

The English had sought by every means to avoid battle throughout their long march from Harfleur and, though accepting it on 25 October as a necessary alternative to capitulation and perhaps lifelong captivity, were finally driven to attack by the pains of hunger and cold. (115)

Keegan's invocation of "the pains of hunger and cold" picks out not physical pains but rather other states that, when intense, are painful. Indeed, English is even messier than this. To Hare's divisions, we might also add a fourth use of "pain" and "hurt" to pick out the *cause* of a pain, either externally ("These tight shoes pain my feet," "Being bitten hurts") or self-generated ("It hurts when I move my arm").

In what follows, I'll reserve "pain" for the first of these uses. Chapter 4 will sort out the distinction between pain and suffering. The first use may actually be the least common one in ordinary discourse. Hare (1964) wryly remarks that it "would be so used more often if the occasions on which we have the sensation without disliking it were not so uncommon" (94). Nevertheless, I take it that once the distinction is made, this usage should be unobjectionable. Which is not to say that the *distinction* is unobjectionable: some readers will think it obvious, and others will remain unconvinced. If you're in the latter category, I offer the distinction provisionally, with the utility of the system to follow as a further argument for adopting it.

Having made the distinction, I'll then argue that pains are exhausted by their imperative content. This will require spelling out the general content of pains (chapter 5) and explaining how pains motivate. Briefly, pains motivate because commands motivate. That is why we give commands, that is how we treat commands, and that's what determines the success conditions for giving and accepting commands. A story about commands should thus shed light on the sense in which pains are intrinsically motivating. Chapter 6 fleshes out that story, showing how pains motivate by giving us reasons for action. I suggest that pains motivate because we accept our bodies as authorities and so as a source of legitimate commands. Pains also differ from one another in a variety of ways. Chapters 7 and 8 show how this variation can be accounted for by the intrinsic content of pains. The virtues of the account may not be obvious. Chapter 9 will defend it against a variety of objections, while chapter 10 will show the advantages of imperativism over accounts that assimilate pains to other intrinsically motivational states.

Imperativism claims that pains are intrinsically motivating. Chapters 11 and 12 will be devoted to dispatching putative counterexamples to this claim. I will show how the pure imperativist can treat otherwise puzzling cases in which people seem to feel pain but are entirely unmoved by it. Spelling out that story will also allow me to make precise the sense in which pains are intrinsically motivating.

An important part of the motivational story will be to distinguish the senses in which pains *don't* motivate intrinsically. Following the distinction made in chapter 4, I'll argue that motivation due to *hurt* or *suffering*—that is, the motivation to remove the pain itself—should not be considered as part of the intrinsic motivational force of pain. I'll further motivate that distinction by showing that it permits a recursive structure that allows for explanation of complex phenomena like the masochistic pleasures (chapter 13). Finally, I'll conclude with a discussion of suffering. In chapter 14, I'll show how imperativism sheds light on the typicality of suffering in response to pain. That is, while pains don't necessarily hurt, they normally do: imperativism can say something about why and give a more sophisticated story than is usually available to those who think that suffering is only contingently related to pain. Thus, while pains promote a particular biological end, they do so in a way that causes more sophisticated creatures like us to suffer.

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